

ADVANCED MACHINE LEARNING · WORKSHEET 12

Regularization (L1 and L2)

Overfitting · Ridge Regression · Lasso Regression

TEACHER COPY — Lecture 12

PART I: THE PROBLEM OF OVERFITTING

Introduction: Why do we need regularization?

▷ HOOK

When a model fits every noisy point, the slope becomes abnormally large, and it fails to generalize. Regularization prevents this by constraining coefficients.

The overfitting problem

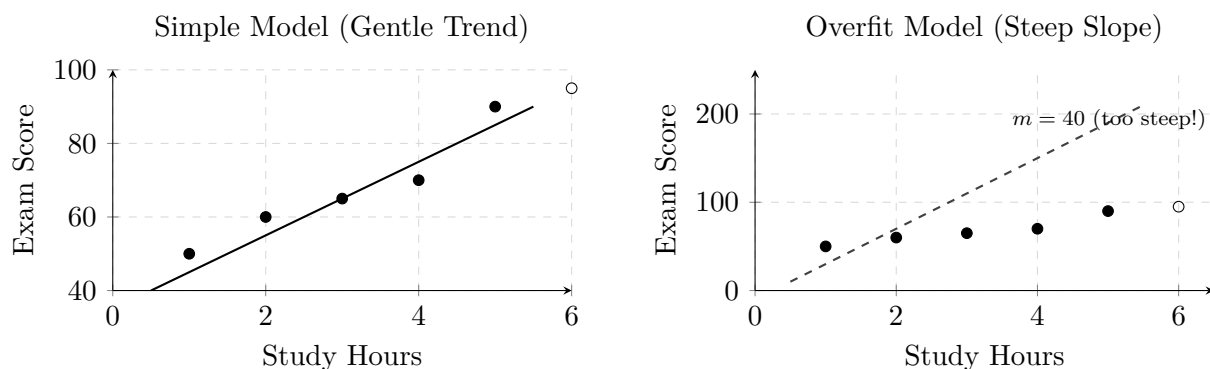
When building machine learning models (Linear Regression, Logistic Regression, etc.), we often see a model perform exceptionally well on training data but fail to generalize to unseen test data. This is called **overfitting**.

An overfitted model has: High **Variance**, Low **Bias**. It memorizes the noise and exact patterns of training data rather than capturing the underlying distribution.

Expected answer: Common Misconception: Students may reverse the terms. Remind them: overfitting = memorizing noise = high variance.

Why fitting every point is dangerous

Quick Example: A simple model draws a gentle trend. An overfit model contorts to hit every point, leading to absurd predictions on new data.



This is why we need a mechanism that says: "Don't try too hard to fit every point."

Geometric intuition in Linear Regression

In 2D space, an overfitted model contorts to pass through every point, becoming hypersensitive to tiny variations in x . The slope m approaches infinity.

Conclusion: Abnormally high coefficient values (steep slopes) are strong indicators of overfitting. A slight change in the test data x will result in a massive change in the prediction $\hat{y}$, leading to high variance and significant test errors.

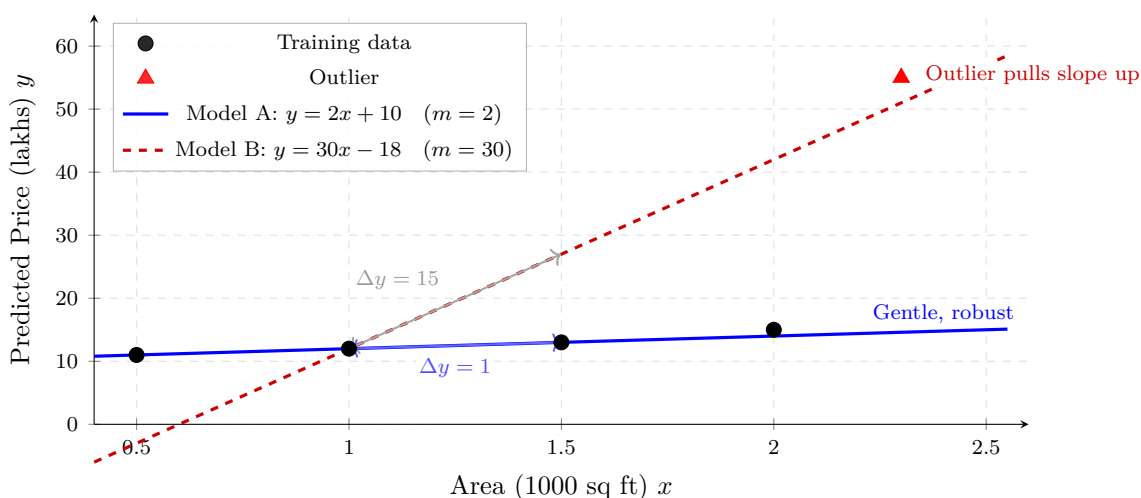
Seeing coefficient explosion in action

Compare two models predicting house price (in lakhs):

	Model A: $y = 2x + 10$	Model B: $y = 30x - 18$
Prediction at $x = 1.0$	$2(1.0) + 10 = 12$	$30(1.0) - 18 = 12$
Prediction at $x = 1.5$	$2(1.5) + 10 = 13$	$30(1.5) - 18 = 27$
Change from $\Delta x = 0.5$	$\Delta y = 1$	$\Delta y = 15$

Model B is highly sensitive, demonstrating high variance due to a large slope.

Comparing Gentle Slope (Model A) vs. Overfitted Slope (Model B)



Question: What does a slope of $m = 4500$ indicate about generalization?

Expected answer: It will perform poorly. A slope of 4500 means that even a tiny change in x (say $\Delta x = 0.01$) causes a prediction change of 45. The model is extremely sensitive — it has memorized the training noise and will produce wildly inconsistent predictions on new data. This is the hallmark of overfitting.

Expected answer: Marking Hint: Full marks for connecting large slope $\rightarrow$ sensitivity $\rightarrow$ high variance $\rightarrow$ poor generalization.

★ KEY INSIGHT

Large coefficient values are the symptom of overfitting. The model is hypersensitive to small changes in input, producing wildly different predictions. We need a mechanism to control these coefficient magnitudes.

What is regularization?

Regularization is a technique used to prevent overfitting by adding a penalty term to the loss function. This penalty discourages the model from using unnecessarily large coefficients.

Regularization is the fundamental approach for controlling parametric models like Linear Regression, Logistic Regression, and Perceptrons.

Types of regularization

Method	Core Mechanism / Description
Ridge Regression (L2 Regularization)	Constrains coefficient magnitudes (prevents exploding slopes)
Lasso Regression (L1 Regularization)	Constrains coefficients and performs automatic feature selection
Elastic Net	Combines both L1 and L2 regularization mechanisms

(Elastic Net is out of scope of this lecture.)

◆ TAKEAWAY

Overfitting shows up as abnormally large coefficients. Regularization prevents this by adding a penalty term to the loss function, forcing the model to balance fitting the data against keeping coefficient magnitudes small.

PART II: L2 REGULARIZATION — RIDGE REGRESSION

Ridge Regression: Geometric and Mathematical Intuition (2D)

▷ HOOK

To control the abnormally high slope m characteristic of an overfitted model, Ridge Regression strategically modifies the standard loss function. Instead of only minimizing the error, the optimizer must also pay a price proportional to the square of the slope.

Modifying the cost function

For Ordinary Least Squares (OLS) linear regression, the cost function is the Sum of Squared Errors (SSE):

$$L_{\text{OLS}} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

We know that abnormally large coefficients cause severe overfitting and high variance. If we want the optimization algorithm to avoid picking large coefficients, what modification should we make to the loss function?

- (A) Subtract the coefficient values from the loss function
- (B) Add a penalty proportional to the squared coefficients to the loss function
- (C) Multiply the loss function by the total number of features

Expected answer: Expected answer: (B) — Adding squared coefficients as a penalty forces the optimizer to keep them small. If a coefficient grows large, its square grows even faster, creating a strong incentive to shrink it. Option (A) would encourage larger coefficients. Option (C) doesn't address coefficient magnitude.

This is exactly what Ridge Regression does.

In Ridge Regression, we append a regularization penalty term:

$$L_{\text{Ridge}} = \sum_{i=1}^n (y_i - mx_i - c)^2 + \lambda m^2$$

Here, $\lambda \geq 0$ is a hyperparameter that controls the intensity of the penalty applied to the slope m .

The mechanism of the penalty (Bias–Variance Tradeoff)

Why does adding λm^2 improve generalization? Let's analyze step-by-step:

1. In an unregularized model ($\lambda = 0$), minimizing training error drives OLS error to 0, but pushes slope m to very large values.
2. In Ridge Regression ($\lambda > 0$), if m becomes large, the penalty term λm^2 explodes, making the overall loss L_{Ridge} high.

Closed-Form Derivation for 2D Ridge Regression

To see exactly how λ controls m , let's derive the slope formula from scratch by minimizing L_{Ridge} :

Step 1: Partial derivative with respect to intercept (c)

Setting $\frac{\partial L_{\text{Ridge}}}{\partial c} = 0$:

$$\sum_{i=1}^n -2(y_i - mx_i - c) = 0 \implies c = \bar{y} - m\bar{x}$$

Step 2: Substitute $c = \bar{y} - m\bar{x}$ back into loss function

Substituting c back into L_{Ridge} eliminates the intercept, expressing the loss purely in terms of slope m :

$$L_{\text{Ridge}}(m) = \sum_{i=1}^n [(y_i - \bar{y}) - m(x_i - \bar{x})]^2 + \lambda m^2$$

Step 3: Differentiate with respect to slope m

Taking the derivative $\frac{dL_{\text{Ridge}}}{dm} = 0$:

$$-2 \sum_{i=1}^n (x_i - \bar{x}) [(y_i - \bar{y}) - m(x_i - \bar{x})] + 2\lambda m = 0$$

Divide by 2 and expand:

$$- \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) + m \sum_{i=1}^n (x_i - \bar{x})^2 + \lambda m = 0$$

Grouping terms with m :

$$m \left(\sum_{i=1}^n (x_i - \bar{x})^2 + \lambda \right) = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

$$m_{\text{Ridge}} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2 + \lambda}$$

Connecting the Derivation to Bias and Variance

Now that we have derived the exact formula $m_{\text{Ridge}} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2 + \lambda}$, we can understand points 3 and 4 with mathematical clarity:

3. **Compromise:** Notice that λ sits directly in the **denominator**. As λ increases, the denominator grows larger while the numerator remains fixed. To minimize L_{Ridge} , the optimizer willingly increases the OLS error (sacrificing a small amount of training accuracy, i.e., introducing slight **Bias**) in order to drastically shrink m .

4. **Result:** A smaller slope m yields a flatter, smoother line. The model becomes far less sensitive to random fluctuations or outliers in training data, drastically reducing **Variance**. This optimizes the Bias–Variance Tradeoff, leading to superior generalization on unseen test data.

Based on the mechanism above, fill in how the compromise works:

What happens	Effect on Bias	Effect on Variance	Net result
m is shrunk toward zero	Increases (slightly — model can't perfectly fit training data)	Decreases (significantly — model is less sensitive to noise)	Better generalization: variance drops more than bias increases

Expected answer: Common Misconception: Students may think both bias and variance decrease. Only variance decreases; bias increases slightly — the net effect is positive because the variance reduction outweighs the bias increase.

Expected answer: Marking Hint: Full marks for correct directions AND the net-effect reasoning.

★ KEY INSIGHT

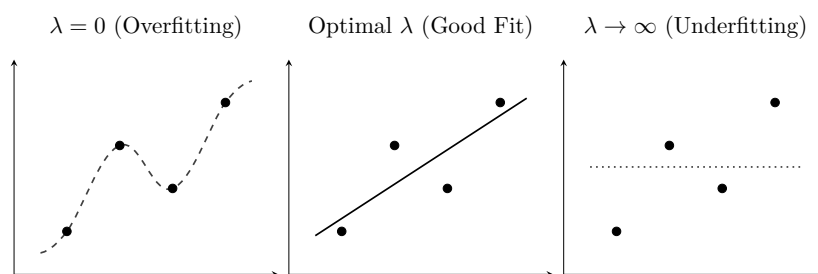
Ridge Regression forces a compromise: it willingly increases bias slightly in order to dramatically reduce variance. The net effect is better generalization to unseen data.

Visualizing the effect of λ

Using a real-world dataset, tuning λ demonstrates three regimes:

λ value	Slope behavior	Regime
$\lambda = 0$	High slope; model = standard OLS	Overfitting
Optimal λ	Moderate slope; main trend captured	Good fit
$\lambda \rightarrow \infty$	$m \approx 0$; essentially flat line	Underfitting

Expected answer: Marking Hint: Award marks for connecting each regime to the bias-variance concept.



◆ TAKEAWAY

Ridge Regression modifies the OLS loss by adding λm^2 . The optimization must compromise between fitting the training data and keeping coefficients small: slight increase in bias, large decrease in variance.

Multiple Ridge Regression (N-Dimensional)

For the general case with p features, the Ridge Normal Equation is:

$$\beta_{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T y$$

(The derivation of this equation is out of scope of this lecture.)

Note: The single-feature (2D) example above is used for mathematical simplicity, while the practical benefits of Ridge Regression become most apparent in multi-feature regression ($N \geq 2$).

The unregularized OLS solution is: $\beta_{\text{OLS}} = (X^T X)^{-1} X^T y$.

Observation: The term λI added to $X^T X$ ensures the matrix is always invertible. This is why Ridge handles **multicollinearity** well — even when $X^T X$ is singular or near-singular, $(X^T X + \lambda I)$ is invertible for any $\lambda > 0$.

◆ TAKEAWAY

In 2D, Ridge adds λ to the denominator of the slope formula, shrinking m toward zero. In N dimensions, the Normal Equation becomes $(X^T X + \lambda I)^{-1} X^T y$, which always exists because λI guarantees invertibility.

Key theoretical properties of Ridge Regression

▷ HOOK

Understanding how Ridge behaves as λ changes is essential for interviews and real applications. These properties form the core of advanced ML understanding.

4.1 The Shrinkage Effect: Do coefficients ever reach absolute zero?

As $\lambda \rightarrow \infty$, what happens to the feature coefficients $(w_1, w_2, \dots)$?

All coefficients shrink asymptotically towards zero. However, in Ridge Regression (L2 penalty), the coefficients will **NEVER** become exactly zero. They become infinitesimally small (e.g., 10^{-5}) but remain non-zero.

Ridge will always use all input features, no matter how heavily regularized it is.

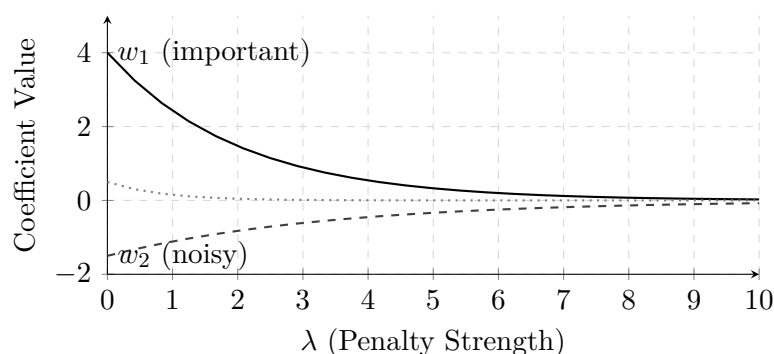
4.2 Asymmetric Shrinkage: Which coefficients shrink faster?

Suppose three features have unregularized OLS coefficients $w_1 = 5000$, $w_2 = 1000$, and $w_3 = 1$. Because the penalty term is λw^2 :

Coefficient	Contribution to penalty
$w_1 = 5000$	$25,000,000 \cdot \lambda$
$w_2 = 1000$	$1,000,000 \cdot \lambda$
$w_3 = 1$	$1 \cdot \lambda$

The larger the initial coefficient, the faster and more aggressively it is penalized. The optimizer immediately targets w_1 for massive reduction. For very high λ , all coefficients converge closely together near zero.

What does this look like visually? If you plot each coefficient's magnitude (y-axis) against λ (x-axis) — called a *regularization path* — the largest coefficients drop steeply first while smaller ones barely move. As λ grows, all curves converge toward zero but never touch the x-axis.

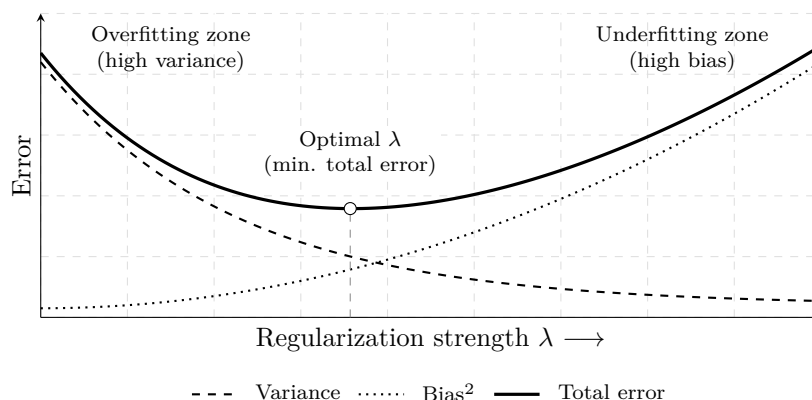


4.3 The Impact on Bias and Variance

Regularization is the direct manipulation of the Bias–Variance Tradeoff.

- **Low λ (e.g., $\lambda \rightarrow 0$):** Coefficients are unrestricted. Bias: Low. Variance: High. The model overfits and fluctuates wildly on test data.
- **High λ (e.g., $\lambda \rightarrow \infty$):** Coefficients are restricted severely, drawing an almost flat line. Bias: High. Variance: Low. The model underfits, basically outputting the mean.

The Sweet Spot: The goal of hyperparameter tuning (via Cross-Validation) is to find the intermediate λ where Variance has dropped significantly but Bias has only increased slightly — minimizing total generalization error (Bias + Variance + Irreducible Error).



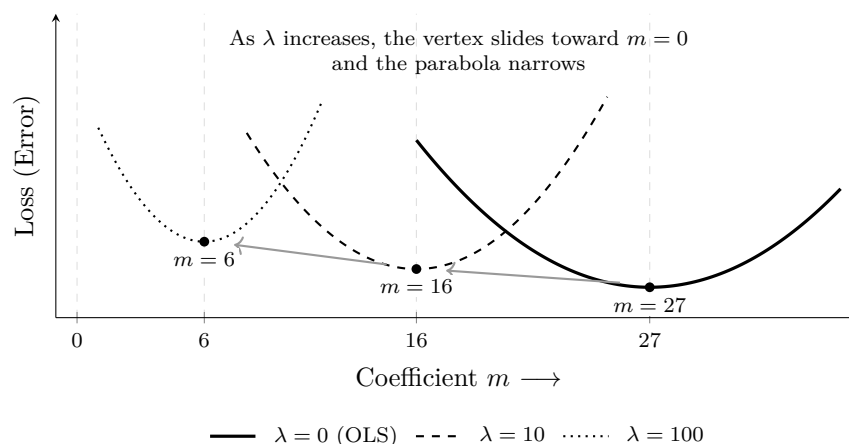
4.4 The Geometric Shifting of the Loss Function

For OLS ($\lambda = 0$), the 2D loss function is a parabola minimized at a large coefficient magnitude.

As λ increases:

1. **The parabola shifts:** The vertex (minimum) moves horizontally from $m = 27$ toward $m = 0$.
2. **The parabola narrows:** The curve becomes steeper or "tighter."
3. **The new minimum:** Because the landscape is translated toward the origin, Gradient Descent converges at a smaller coefficient value.

For a 3D contour plot with two weights (w_1, w_2), the center of the elliptical contours shifts diagonally toward the origin $(0, 0)$ as λ increases.



4.5 Why is it called "Ridge" Regression? (Hard Constraint Intuition)

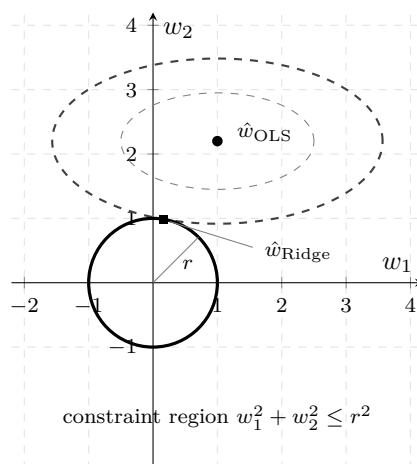
Note: This section is optional self-reading for interested students. It will not be covered in the lecture and is not part of the examination.

Until now, we have viewed Ridge as a **Soft Constraint** — adding a penalty directly to the loss. Mathematically, this is equivalent to solving OLS subject to a **Hard Constraint**:

$$\text{Minimize } (XW - Y)^T(XW - Y) \quad \text{subject to} \quad w_1^2 + w_2^2 + \dots + w_n^2 \leq r^2$$

The geometric picture:

- In a 2D parameter space (w_1, w_2) , the constraint $w_1^2 + w_2^2 \leq r^2$ forms a perfect **circle** centered at the origin. The radius r is inversely proportional to λ .
- The OLS loss function forms **elliptical** contour lines expanding outward from its global minimum (which usually sits far outside the circle).
- **The intersection:** The solution lies on the **boundary (perimeter)** of the constraint circle where the OLS contour ellipse barely touches it (tangent point).



Because the solution is forced to lie on the boundary or "**ridge**" of this constraint circle, the technique earned the name **Ridge Regression**.

4.6 A Practical Heuristic

Ridge Regression becomes genuinely useful when the number of input features is $N \geq 2$. With only one feature (simple linear regression), applying Ridge is rarely beneficial. The true power of L2 regularization is managing the complex interplay, multicollinearity, and overfitting potential in high-dimensional, multi-feature spaces.

★ KEY INSIGHT

Ridge shrinks all coefficients toward zero but never makes them exactly zero. Larger coefficients are penalized more aggressively. The constraint region is a circle, and the name "Ridge" comes from the solution lying on the boundary (ridge) of that circle.

◆ TAKEAWAY

Six key properties: (1) Coefficients never reach exactly zero. (2) Larger coefficients shrink faster. (3) Bias increases and variance decreases with λ . (4) The loss function parabola shifts toward the origin. (5) The circle constraint explains the name "Ridge." (6) Ridge is most useful with $N \geq 2$ features.

PRACTICE P1 · RIDGE REGRESSION

(a) A student trains a Linear Regression model on a housing dataset with 30 features. The training RMSE is 1,200 but the test RMSE is 15,800. The top three coefficients are $w_5 = 8400$, $w_{12} = -6200$, and $w_{23} = 5100$.

(i) Is this model likely overfitting or underfitting? What evidence supports your answer?

Overfitting. Evidence: (1) Huge gap between training RMSE (1,200) and test RMSE (15,800) — the model fits training data well but fails on new data. (2) Abnormally large coefficients (8400, -6200, 5100) indicate the model is hypersensitive to small input changes. Both are classic overfitting symptoms.

(ii) If Ridge Regression is applied with a well-tuned λ , what do you expect to happen to the training RMSE and the test RMSE?

Training RMSE will increase (because Ridge introduces bias — the model can no longer perfectly chase training points). **Test RMSE will decrease significantly** (because Ridge reduces variance — the model generalizes better). The gap between the two will shrink.

(iii) Will the coefficients w_5, w_{12}, w_{23} remain at their current values, increase, or decrease in magnitude after Ridge is applied?

Decrease in magnitude. Ridge penalizes λw^2 , which is enormous for coefficients like 8400. The optimizer will aggressively shrink these large coefficients. In fact, by asymmetric shrinkage (Section 4.2), the largest coefficient ($w_5 = 8400$) will be shrunk the most.

Expected answer: Common Misconception: Students may think Ridge reduces both training and test error. It typically increases training error while decreasing test error.

Expected answer: Marking Hint: Full marks for correct direction of both RMSEs and the asymmetric shrinkage connection.

(b) Given:

$$X^T X = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}, \quad X^T y = \begin{bmatrix} 10 \\ 6 \end{bmatrix}, \quad \lambda = 2$$

(i) Write and compute $(X^T X + \lambda I)$:

$$\begin{bmatrix} 4+2 & 2 \\ 2 & 4+2 \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 2 & 6 \end{bmatrix}$$

(ii) Compute $\beta_{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T y$:

$$\det = 6 \times 6 - 2 \times 2 = 32$$

$$(X^T X + \lambda I)^{-1} = \frac{1}{32} \begin{bmatrix} 6 & -2 \\ -2 & 6 \end{bmatrix}$$

$$\beta_{\text{ridge}} = \frac{1}{32} \begin{bmatrix} 6 & -2 \\ -2 & 6 \end{bmatrix} \begin{bmatrix} 10 \\ 6 \end{bmatrix} = \frac{1}{32} \begin{bmatrix} 48 \\ 16 \end{bmatrix} = \begin{bmatrix} 1.5 \\ 0.5 \end{bmatrix}$$

(iii) Now compute $\beta_{\text{OLS}} = (X^\top X)^{-1} X^\top y$:
 $\det = 4 \times 4 - 2 \times 2 = 12$

$$\beta_{\text{OLS}} = \frac{1}{12} \begin{bmatrix} 4 & -2 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 10 \\ 6 \end{bmatrix} = \frac{1}{12} \begin{bmatrix} 28 \\ 4 \end{bmatrix} = \begin{bmatrix} 2.33 \\ 0.33 \end{bmatrix}$$

(iv) Compare: which set of coefficients has a smaller total squared magnitude ($\sum \beta_j^2$)?

Ridge: $1.5^2 + 0.5^2 = 2.25 + 0.25 = 2.50$. **OLS:** $2.33^2 + 0.33^2 \approx 5.43 + 0.11 = 5.54$. **Ridge has smaller total squared magnitude (2.50 vs 5.54). This makes sense because Ridge explicitly minimizes $\sum \beta_j^2$ as part of its objective.**

Expected answer: Marking Hint: Award marks for each computation step. Part (iv) requires comparison and reasoning.

PRACTICE P1-B · RIDGE REGRESSION (Homework)

(a) A model has OLS coefficients $w_1 = 200$ and $w_2 = 3$. You apply Ridge with a moderate λ .

(i) Which coefficient will shrink more in absolute terms? Explain using the penalty mechanism.
 w_1 will shrink more in absolute terms. The penalty is λw^2 : w_1 contributes $200^2 \cdot \lambda = 40,000\lambda$ while w_2 contributes $3^2 \cdot \lambda = 9\lambda$. The optimizer gets a much larger loss reduction by shrinking w_1 than w_2 .

(ii) After regularization, could w_2 end up larger than w_1 ? Explain.

Theoretically possible if λ is large enough. Ridge shrinks all coefficients proportionally to their magnitude. With very high λ , both approach zero, and w_1 is shrunk more aggressively. However, in practice, the relative ordering usually remains the same unless λ is extreme. The key point is that the larger coefficient is always penalized more.

Expected answer: Marking Hint: Part (i) requires the w^2 scaling argument. Part (ii) is conceptual — accept reasonable answers.

(d) **Concept check: The denominator argument.** The Ridge formula for a 2D slope is:

$$m_{\text{Ridge}} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2 + \lambda}$$

A student says: "If I set $\lambda = 10^{12}$, the coefficient will be exactly zero." Is this correct? Why or why not?

Incorrect. With $\lambda = 10^{12}$, the result is $\frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{10^{12} + \sum (x_i - \bar{x})^2}$, which is an extremely small number (e.g., 10^{-10}) but NOT exactly zero. Division by a finite number never gives exactly zero unless the numerator is exactly zero. This is the fundamental reason why Ridge coefficients never reach absolute zero — λ is in the denominator, creating asymptotic shrinkage, not exact zeroing.

Expected answer: Common Misconception: Students may think "practically zero" is the same as "exactly zero." Mathematically, they are different, and this difference is why Ridge cannot perform feature selection.

Expected answer: Marking Hint: Full marks for the division-never-gives-zero argument and connecting it to the denominator position.

PART III: L1 REGULARIZATION — LASSO REGRESSION

Lasso Regression: L1 Regularization

▷ HOOK

Ridge Regression pulls coefficients closer to zero but never exactly makes them zero. What if we want a model that not only controls coefficients but also identifies which features are truly important and automatically discards the rest?

A quick analogy

Ridge is like turning down the volume on every speaker in a room — everyone still speaks, but more softly. No one is removed. **Lasso** is like a talent show judge — only the best performers stay on stage. Everyone else is sent home (coefficient = 0).

Keep this analogy in mind as we formalize Lasso mathematically.

The mathematical formulation

Lasso (Least Absolute Shrinkage and Selection Operator) uses a different penalty. Instead of the squared magnitude, Lasso penalizes the absolute magnitude of the coefficients.

$$L_{\text{Lasso}} = \underbrace{\sum_{i=1}^n (y_i - \hat{y}_i)^2}_{\text{OLS error}} + \lambda \underbrace{\sum_{j=1}^p |w_j|}_{\text{L1 penalty}}$$

Here:

- w_j : the coefficients (slopes) of the input features
- λ (or α): the hyperparameter controlling the strength of regularization, $\lambda \geq 0$

The impact of increasing λ

λ value	Coefficient behavior	Model state
$\lambda = 0$	Standard OLS values; potentially large and overfitting	Overfitting
Intermediate λ	Less important features hit exactly zero; remaining shrink	"Sweet spot" for feature selection
Very high λ	All coefficients forced to zero; flat line ($\hat{y} = c$)	Underfitting

◆ TAKEAWAY

Lasso uses L1 penalty ($\lambda \sum |w_j|$). The critical behavioral difference from Ridge emerges in the next section.

Key Differences: Lasso vs. Ridge Regression

▷ HOOK

The change from w_j^2 (L2 norm) to $|w_j|$ (L1 norm) might seem minor mathematically, but it leads to profoundly different behavior in the resulting models.

Feature Selection and Sparsity

The most critical difference is that **Lasso Regression can drive the coefficients of less important features exactly to zero**.

- **Ridge Regression:** Shrinks coefficients, keeping all features in the model (e.g., $w_1 = 0.5, w_2 = 0.001$).
- **Lasso Regression:** Acts as an automatic **Feature Selection** mechanism. If a feature does not significantly contribute to predicting the target variable y , Lasso forces its coefficient to 0.

When multiple coefficients become exactly zero, the resulting weight vector is called **sparse**. Lasso Regression is famous for creating **sparsity** in models.

When to use Lasso vs. Ridge?

Scenario	Preferred method	Reason
High-dimensional data with many irrelevant features	Lasso	Drops irrelevant columns, simplifies model, reduces dimensionality, easier to interpret
Domain knowledge says all features contribute	Ridge	Keeps all features with reduced influence, preserves all information

★ KEY INSIGHT

Ridge keeps all features and only reduces their magnitudes. Lasso eliminates features entirely. This makes Lasso a dimensionality reduction tool and Ridge a coefficient stabilization tool.

◆ TAKEAWAY

Ridge and Lasso represent different beliefs about the coefficient structure. Ridge assumes all features contribute (dense model). Lasso assumes many features are irrelevant (sparse model). Neither is universally better — the choice depends on the problem.

Optimization note: Ridge has a closed-form solution (one-step matrix computation), while Lasso requires iterative numerical optimization (e.g., coordinate descent) because the L1 penalty creates a non-differentiable kink at zero.

Geometric and Visual Intuition of Lasso

▷ HOOK

We now understand what Lasso does differently. But why does it happen? The answer lies in the geometry of the L1 penalty — both in the shape of the loss function and the shape of the constraint region.

Geometric interpretation: Diamond vs. Circle

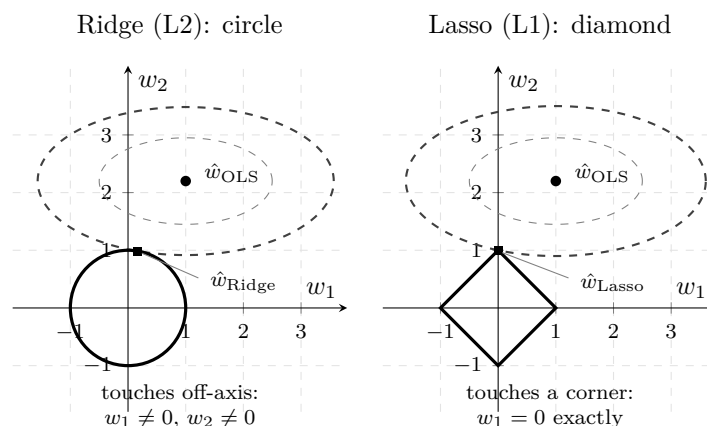
Note: This section is optional extra reading for interested students. It will not be covered in the lecture and is not part of the examination.

Lasso constrains the coefficients to lie within:

$$\sum_{j=1}^p |w_j| \leq t$$

In two dimensions, this is a **diamond** (rotated square) centered at the origin.

Method	Constraint shape	Why the shape matters
Ridge	Circle (smooth boundary)	RSS contours touch the smooth surface at a non-axis point — no coefficient is forced to zero
Lasso	Diamond (sharp corners on axes)	RSS contours are likely to first touch at a corner — where one or more coefficients are exactly zero



Bias–Variance tradeoff in Lasso

Just like Ridge, Lasso modulates the Bias–Variance Tradeoff:

- At **low** λ : model is complex. Low Bias, High Variance (overfitting).
- As λ **increases**: penalty introduced. Variance drops significantly, Bias slowly climbs.
- At **very high** λ : all weights become zero. Extremely high Bias (underfitting).

The goal is to use cross-validation to select a λ in the middle zone.

◆ TAKEAWAY

Lasso's L1 penalty creates a "V"-shaped loss at $w = 0$ and a diamond-shaped constraint region. The diamond's sharp corners on the axes cause RSS contours to touch at vertices, forcing coefficients to exactly zero.

Why does Lasso create sparsity? (Mathematical proof)

▷ HOOK

A classic ML interview question: "Why does Lasso (L1) force coefficients exactly to zero, while Ridge (L2) only pushes them close to zero?" The answer lies in where λ appears in the coefficient formula.

The derivative of the Lasso loss function

Consider a single feature x , slope m . The Lasso loss is:

$$L_{\text{Lasso}} = \sum_{i=1}^n (y_i - mx_i - c)^2 + \lambda|m|$$

Because $|m|$ is not differentiable at $m = 0$, we analyze using a piecewise approach.

Case 1: When $m > 0$

$|m| = m$. Penalty = λm . Taking the derivative and setting to zero:

$$m = \frac{\text{OLS numerator} - \lambda}{\text{denominator}}$$

λ appears in the numerator as a **direct subtraction**.

- Suppose OLS numerator = 100.
- If $\lambda = 50$: numerator = $100 - 50 = 50 > 0 \rightarrow m > 0$ ✓
- If $\lambda = 100$: numerator = $0 \rightarrow m = 0$
- If $\lambda = 120$: formula gives negative, but we assumed $m > 0$. Algorithm halts at $m = 0$.

Case 2: When $m < 0$

$|m| = -m$. Penalty = $-\lambda m$. The derivative yields:

$$m = \frac{\text{OLS numerator} + \lambda}{\text{denominator}}$$

λ is **added** to the numerator. As λ increases, a negative m is driven toward zero. Once the formula outputs a positive number, the assumption is violated, and the algorithm clips at $m = 0$.

Expected answer: Common Misconception: Students may think Lasso uses a different base formula. The OLS part is the same; only the penalty derivative changes.

Expected answer: Marking Hint: Full marks require understanding both cases and why the coefficient is clipped to zero.

Comparing with Ridge Regression's formula

In Ridge, the derivative yields:

$$m = \frac{\text{OLS numerator}}{\text{denominator} + \lambda}$$

λ sits in the **denominator**. No matter how large λ becomes (e.g., 10^9), you are dividing the numerator by a massive number. The result becomes infinitesimally small (e.g., 10^{-9}) but mathematically **never** reaches exactly 0.

★ KEY INSIGHT

- **Lasso (L1):** λ in the **numerator** (subtraction) $\rightarrow$ coefficient is forced to exactly zero once λ surpasses the feature's correlation with the target.
- **Ridge (L2):** λ in the **denominator** (division) $\rightarrow$ coefficient shrinks asymptotically but never reaches exactly zero.

◆ TAKEAWAY

L1 creates sparsity because λ subtracts directly in the numerator. L2 does not create sparsity because λ only scales the denominator. This is the mathematical reason why Lasso performs feature selection and Ridge does not.

PRACTICE P2 · LASSO, SPARSITY, AND COMPARISON

(a) Application: Feature importance. A hospital uses 50 features to predict patient readmission. After applying Lasso with cross-validated λ , 42 coefficients become zero and only 8 survive.

(i) What term describes a model with many zero coefficients?

Sparse (or "sparse model").

(ii) A colleague suggests using Ridge instead "because it is more stable." If Ridge were applied,

approximately how many features would remain in the model? Explain why.

All 50 features would remain. Ridge never sets coefficients exactly to zero — it shrinks them toward zero but keeps all features. This is because λ appears in the denominator of the Ridge formula, causing asymptotic shrinkage but never exact zeroing.

(iii) From a clinical interpretability standpoint, which model (the 8-feature Lasso or the 50-feature Ridge) would be easier for doctors to explain to patients? Why?

The 8-feature Lasso model would be easier. Doctors can say: "Your readmission risk is driven by these 8 factors." With 50 features, explanations become impractical. Lasso's automatic feature selection creates a simpler, more interpretable model.

Expected answer: Marking Hint: Full marks for correct feature count (50), correct reason (denominator argument), and interpretability reasoning.

(b) Concept check: The numerator/denominator argument. Consider the Lasso coefficient formula: $m = \frac{\text{OLS numerator} - \lambda}{\text{denominator}}$ (for $m > 0$).

Suppose OLS numerator = 80 and denominator = 40.

(i) For $\lambda = 30$, compute m : $m = \frac{80-30}{40} = \frac{50}{40} = 1.25$

(ii) For $\lambda = 80$, compute m : $m = \frac{80-80}{40} = \frac{0}{40} = 0$

(iii) For $\lambda = 120$, what does the formula give? $\frac{80-120}{40} = \frac{-40}{40} = -1$. What does the algorithm actually do? **Clips to $m = 0$ because the $m > 0$ assumption is violated.**

Now consider the Ridge formula: $m = \frac{80}{40+\lambda}$.

(iv) For $\lambda = 10^6$, compute m : $m = \frac{80}{40+10^6} = \frac{80}{1,000,040} \approx 0.00008$. Is it exactly zero? **No — it is extremely small but not zero. This is the denominator argument in action.**

Expected answer: Marking Hint: Full marks for all computations and the key observation: Lasso hits exact zero, Ridge does not.

(c) Application: Choosing the right method. For each scenario, choose Ridge or Lasso and give one reason.

Scenario	Choice	Reason
Gene expression data with 10,000 genes, most expected to be irrelevant	Lasso	Most genes are irrelevant; Lasso drops them, creating a sparse model and reducing dimensionality
Predicting sales using 5 well-understood economic indicators	Ridge	All 5 features are known to matter; Ridge preserves all features with stable coefficients
20 highly correlated sensor readings from a factory	Ridge	Ridge handles multicollinearity well (λI stabilizes $X^\top X$). Lasso may arbitrarily pick one from correlated groups
Customer churn model with 200 features; business wants a simple scorecard	Lasso	Business needs interpretability. Lasso selects the most important features, creating a simple scorecard

Expected answer: Common Misconception: For the sensor scenario, students may choose Lasso thinking "remove duplicates." But Lasso is unstable with correlated features — Ridge is better for stabilization.

Expected answer: Marking Hint: Award marks for correct choice AND relevant reasoning. The sensor question is the hardest.

(d) **Geometric reasoning.** A student draws two constraint regions on the same axes — a circle (Ridge) and a diamond (Lasso). The OLS solution is at $(3, 0.5)$.

(i) Which constraint region is more likely to produce a solution where $w_2 = 0$?

The Lasso diamond.

(ii) Explain geometrically why:

The diamond has sharp corners on the w_1 axis (where $w_2 = 0$) and the w_2 axis (where $w_1 = 0$). The elliptical RSS contours centered near $(3, 0.5)$ are more likely to first touch the diamond at one of these corners — where a coefficient is zero. The circle has no corners on the axes, so the tangent point is generally at a position where both w_1 and w_2 are non-zero.

Expected answer: Marking Hint: Full marks require mentioning corners on axes, tangent point, and why the circle avoids axis solutions.

(e) **Penalty calculation.** Given the coefficient vector $\mathbf{w} = [5, -2, 8]$, compute:

(i) The L1 penalty ($\sum |w_j|$):

$$|5| + |-2| + |8| = 5 + 2 + 8 = 15$$

(ii) The L2 penalty ($\sum w_j^2$):

$$5^2 + (-2)^2 + 8^2 = 25 + 4 + 64 = 93$$

(iii) Which penalty would a Lasso model use? **L1 (sum of absolute values = 15).** Which would Ridge use? **L2 (sum of squares = 93).**

Expected answer: Common Misconception: Students often confuse absolute value with squaring for negative numbers. $|-2| = 2$ but $(-2)^2 = 4$ — different operations, different penalties, different behavior.

Expected answer: Marking Hint: Full marks for correct computation of both penalties and correct assignment to Ridge/Lasso.

PART IV: SIDE-BY-SIDE COMPARISON AND PRACTICAL CONSIDERATIONS

Complete comparison and practical usage

Side-by-side comparison

Aspect	Ridge (L2)	Lasso (L1)
Penalty term	$\lambda \sum w_j^2$	$\lambda \sum w_j $
Shrinkage	Toward zero, never exactly zero	Some coefficients become exactly zero
Feature selection	No — keeps all features	Yes — automatic feature selection
λ position in formula	Denominator (scaling)	Numerator (subtraction)
Constraint geometry	Circle (smooth)	Diamond (sharp corners)
Loss function shape	Smooth parabola	"V"-shaped kink at zero
Closed-form solution	Yes: $(X^\top X + \lambda I)^{-1} X^\top y$	No — requires numerical optimization
Best when	All features contribute; multi-collinearity	Many features irrelevant; want interpretable model

Cross-validation for λ selection

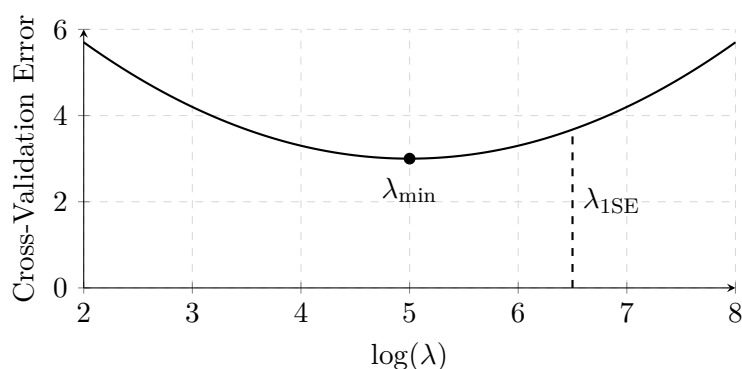
To find the optimal regularization strength λ :

1. Evaluate a candidate grid of λ values using k -fold cross-validation.
2. Select $\lambda_{\min}$ (lowest validation error) or λ_{1SE} (simplest model within 1 std error of minimum).

Classroom Exercise: A team tests four λ values. Which should they select?

λ	0	0.1	1	10
Avg. Validation MSE	150	85	72	310

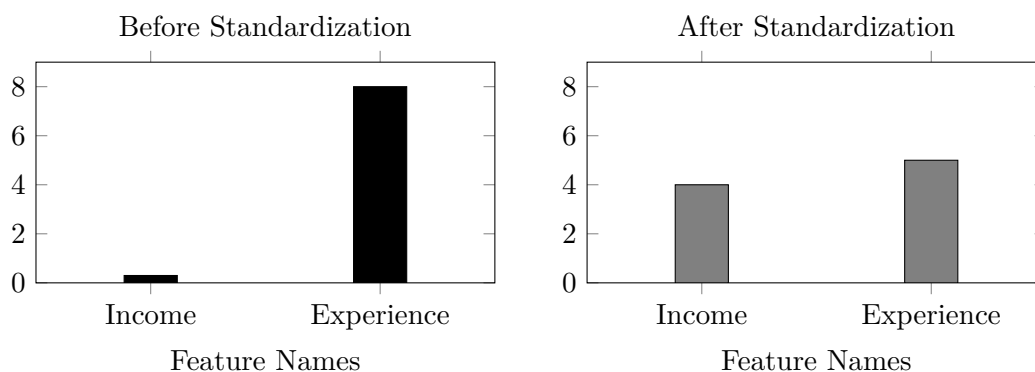
Answer: $\lambda = 1$, because it has the lowest average validation MSE (72). $\lambda = 0$ overfits (high validation error), and $\lambda = 10$ underfits (very high error). This confirms the U-shaped behavior.



Standardization before regularization

Because regularization penalizes raw coefficient magnitudes (w_j), features on smaller scales have naturally larger coefficients and get penalized far more heavily.

Rule: Standardize all features ($x_j^{\text{std}} = \frac{x_j - \mu_j}{\sigma_j}$) before fitting. Do **not** regularize the intercept term (β_0).



★ KEY INSIGHT

Standardization ensures the penalty treats all features equally regardless of measurement scales. Without it, a feature in thousands (income in rupees) would be penalized far more than one in single digits (years of experience).

◆ TAKEAWAY

Ridge keeps all features with reduced coefficients; Lasso eliminates unimportant ones. λ is selected via cross-validation. Features must be standardized so the penalty is fair. The intercept is not regularized.

Common Mistakes to Avoid

Mistake	Why it's wrong
Regularizing the intercept (β_0)	The intercept shifts the prediction baseline — penalizing it biases the model toward $\hat{y} = 0$ instead of $\hat{y} = \bar{y}$
Standardizing before train-test split	This leaks test-set statistics into training, producing overly optimistic results (data leakage)
Assuming larger λ is always better	Too large λ causes underfitting — the optimal λ is found via cross-validation
Believing Ridge performs feature selection	Ridge shrinks but never zeros out coefficients — only Lasso performs feature selection
Using raw (unstandardized) features	The penalty unfairly targets features with naturally large coefficients due to measurement scale

Interview Traps — True or False?

Statement	Answer
"Ridge Regression performs feature selection."	False. Ridge shrinks toward zero but never makes coefficients exactly zero. Only Lasso performs feature selection.
"Ridge coefficients can become exactly zero if λ is large enough."	False. λ is in the denominator ($m = \frac{\text{num}}{\text{denom} + \lambda}$), so the result is asymptotically small but never exactly zero.
"Lasso always outperforms Ridge."	False. When all features are relevant or highly correlated, Ridge often outperforms Lasso. Neither is universally better.
"The intercept (β_0) should be regularized along with other coefficients."	False. The intercept captures the baseline prediction ($\bar{y}$). Regularizing it biases predictions toward zero.

PRACTICE P3 · APPLICATION AND SYNTHESIS

(a) **Debugging a pipeline.** A data science team applies Lasso to predict crop yield using 4 features:

- Rainfall (mm, range 200–2000)
- Temperature ($^{\circ}\text{C}$, range 10–45)
- Soil pH (range 4.5–8.0)
- Fertilizer cost (Rs., range 5000–50000)

After training, Lasso sets the coefficients for Rainfall and Soil pH to zero, keeping only Temperature and Fertilizer cost. A domain expert is surprised: "Rainfall is definitely important for crop yield!"

(i) What preprocessing step did the team most likely skip?

Standardization. The features are on vastly different scales.

(ii) Why would skipping this step cause Rainfall's coefficient to be unfairly treated by the penalty? **Rainfall has large values (200–2000) $\rightarrow$ small coefficient $\rightarrow$ penalty crushes it easily. Without standardization, the penalty targets coefficient magnitude regardless of feature importance.**

(iii) After fixing the issue, would you expect the same features to be eliminated? Explain.

No. After standardization, all features are on the same scale (mean 0, std 1). The penalty now fairly compares feature importance rather than being biased by measurement units. If Rainfall is truly important for crop yield (as the domain expert says), its standardized coefficient should be large enough to survive the Lasso penalty.

Expected answer: Common Misconception: Students may think "Lasso correctly identified irrelevant features." The elimination was an artifact of scale differences, not feature importance.

Expected answer: Marking Hint: Full marks for identifying standardization, explaining the scale-coefficient relationship, and predicting the post-fix outcome.

(b) **Interpreting regularization paths.** Below are two regularization path descriptions.

Path A: As λ increases from 0 to 100, all 10 coefficient curves smoothly approach zero but none touches the x-axis.

Path B: As λ increases from 0 to 100, coefficient curves shrink and hit zero one by one. At $\lambda = 100$, only 3 features remain non-zero.

(i) Which path corresponds to Ridge? **Path A.** Which to Lasso? **Path B.**

(ii) At $\lambda = 100$, how many features does each model effectively use?

Path A: 10 (all features — Ridge never zeros out coefficients). Path B: 3 (only 3 remain non-zero — Lasso eliminated 7 features).

Expected answer: Marking Hint: Award marks for correct identification and correct feature

counts with reasoning.

(c) **Sequence arrangement.** A colleague sends you an unordered list of steps for building a Ridge model. Put them in order.

Correct order:

1. Split data into training and test sets
2. Fit mean μ_j and std σ_j on training set only
3. Standardize features using training-set statistics
4. Select λ via cross-validation
5. Compute $(X^\top X + \lambda I)^{-1} X^\top y$
6. Evaluate on the held-out test set

Why must you fit μ_j and σ_j only on the training set?

To prevent data leakage. If we compute statistics using the entire dataset (including test data), the model indirectly "sees" test information during training, producing overly optimistic performance estimates. The test set must remain completely unseen during all training and preprocessing steps.

Expected answer: Common Misconception: Students may standardize before splitting, causing data leakage.

Expected answer: Marking Hint: Full marks for correct ordering AND the data leakage explanation.

(d) **Interview question: Why is the intercept not regularized?**

In the Ridge loss function, the penalty is $\lambda \sum_{j=1}^p w_j^2$ — the summation starts at $j = 1$, deliberately excluding the intercept w_0 (or β_0). Why?

The intercept represents the baseline prediction — it equals $\bar{y}$ when all features are centered. If we regularize it, we push the baseline toward zero instead of toward $\bar{y}$, which would systematically bias every prediction. The intercept doesn't contribute to model complexity or overfitting — it simply shifts the prediction up or down. Penalizing it harms the model without any benefit.

Expected answer: Marking Hint: Full marks for explaining that the intercept captures the baseline prediction and is unrelated to model complexity.

PRACTICE P4 · COMPREHENSIVE CHALLENGE (Homework)

(a) **Case study.** An e-commerce company wants to predict customer lifetime value (CLV) using 80 features. The data scientists know from experience that only about 10–15 features actually drive CLV. The remaining features are mostly marketing campaign IDs, timestamp encodings, and other noisy variables.

(i) Should they use Ridge or Lasso? Justify.

Lasso. The team suspects only 10–15 features matter out of 80. Lasso will automatically identify and keep the important features while forcing the irrelevant ones (campaign IDs, timestamps, noise) to exactly zero. This produces a simpler, more interpretable model.

(ii) After applying Lasso, training accuracy is 97% and test accuracy is 94%. The λ was selected via cross-validation. Is this a good model? Explain.

Yes, this appears to be a good model. The gap between training (97%) and test (94%) accuracy is only 3 percentage points — a small gap suggests the model is not severely overfitting. The λ was properly selected via cross-validation rather than being manually set. The slight drop from training to test is expected (the model always fits training data slightly better).

(iii) A manager asks: "Why not just set $\lambda = 0$ to get the best training accuracy?" Write a response using the Bias–Variance Tradeoff.

Setting $\lambda = 0$ removes all regularization, giving the model full freedom to fit training

data. While training accuracy may reach 99-100%, the model will overfit — it memorizes noise and specific patterns in the training data. The test accuracy will drop significantly because the model does not generalize. Regularization ($\lambda > 0$) introduces a small amount of bias (slightly lower training accuracy) but dramatically reduces variance (much better test accuracy). The cross-validated λ finds the optimal compromise where total generalization error is minimized.

Expected answer: Marking Hint: Full marks require connecting $\lambda = 0$ to overfitting, training vs test gap, and the bias-variance compromise.

(b) **Error identification.** A student writes the following Ridge Regression cost function:

$$J(\beta) = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=0}^p |\beta_j|$$

Identify and explain all errors (there are at least two).

Error 1: The student claims this is Ridge but uses $|\beta_j|$ (absolute value = L1 = Lasso penalty) instead of β_j^2 (squared = L2 = Ridge penalty).

Error 2: The summation starts at $j = 0$, which includes the intercept β_0 in the penalty. The intercept should NOT be regularized — the summation should start at $j = 1$.

Expected answer: Marking Hint: Award marks for each error identified with clear explanation.

(c) **Conceptual scenario.** Two colleagues are debating:

Colleague A: "I used Ridge with $\lambda = 1000$ and all my coefficients are zero. Ridge can produce zero coefficients if λ is large enough."

Colleague B: "That's impossible. Ridge coefficients never become exactly zero no matter how large λ is."

Who is correct? Explain using the mathematical formula.

Colleague B is correct. In the Ridge formula, $m = \frac{\text{numerator}}{\text{denominator} + \lambda}$, the coefficient is obtained by dividing by (denominator + λ). Even with $\lambda = 1000$ or $\lambda = 10^{15}$, the result is a very small non-zero number. Division by a finite denominator never produces exactly zero (unless the numerator is exactly zero, which would be an exceptional coincidence). Colleague A's coefficients are likely extremely small (e.g., 10^{-6}) and appear as zero due to rounding in the software display, but mathematically they are not exactly zero.

Expected answer: Marking Hint: Full marks for identifying B as correct, the denominator argument, and the rounding explanation.

(d) **Synthesis questions.** Answer any three.

(i) Geometrically, why does Lasso produce zero coefficients but Ridge does not?

Ridge constrains coefficients to a circle (smooth boundary). RSS contours touch the smooth circle at a point that is generally not on an axis — both coefficients remain non-zero. Lasso constrains to a diamond with sharp corners on the coordinate axes. RSS contours are more likely to touch at a corner — where one or more coefficients are exactly zero.

(ii) Mathematically, why does L1 create sparsity but L2 does not?

In L1, λ appears in the numerator as a subtraction: $m = \frac{\text{OLS} - \lambda}{\text{denom}}$. Once λ exceeds the OLS value, the coefficient is clipped to exactly zero. In L2, λ appears in the denominator: $m = \frac{\text{OLS}}{\text{denom} + \lambda}$. Division by a large number produces a tiny but non-zero value.

(iii) A model uses Lasso and sets λ very large. Describe the resulting model.

Bias: Very high (flat line). **Variance:** Very low. **Features used:** Zero — all forced to zero. Model outputs only the intercept. Extreme underfitting.

(iv) How does λ relate to the bias-variance tradeoff?

$\lambda = 0$: No penalty, OLS, low bias, high variance (overfitting). Optimal λ : Slight bias increase, large variance decrease. $\lambda \rightarrow \infty$: High bias, low variance (underfitting).

Optimal λ minimizes total error via cross-validation.

(v) Why must features be standardized before regularization?

Coefficient magnitude depends on feature scale. Without standardization, the penalty unfairly targets large-coefficient features. Standardization ($x_j^{\text{std}} = \frac{x_j - \mu_j}{\sigma_j}$) puts all features on the same scale, making the penalty fair.

(vi) A company has 15 features: 5 are known to be highly predictive and the rest are noisy. Recommend a method.

Lasso. Since only 5 out of 15 features are important, Lasso will force the 10 noisy features' coefficients to zero, keeping only the 5 that matter. This creates a simpler, interpretable model. Ridge would keep all 15 features — the noisy ones would have small but non-zero coefficients, adding unnecessary complexity.

Expected answer: Marking Hint: Award 3 marks per question (any three). Full marks require both the mechanism and the conclusion.

(e) **Evaluate: Defend your recommendation.** A pharmaceutical company is building a model to predict drug efficacy using 200 molecular features. The business team requires an interpretable model — regulators need to know *which* molecular properties drive efficacy, and the model must be explainable in a submission report.

Should the team use Ridge or Lasso? Defend your choice by addressing: (1) interpretability, (2) number of features, and (3) regulatory requirements.

Lasso. (1) **Interpretability:** Lasso automatically selects only the most important molecular features by forcing the rest to zero — "these 12 out of 200 properties drive efficacy." Ridge keeps all 200 with tiny coefficients, making interpretation impractical. (2) **Number of features:** With 200 features, many are likely irrelevant. Lasso's sparsity is ideal for dimensionality reduction. (3) **Regulatory requirements:** A sparse Lasso model with 10–20 features is far more auditable and defensible than a dense Ridge model with 200 non-zero coefficients.

Expected answer: Marking Hint: This is a Bloom Level "Evaluate" question. Full marks require weighing trade-offs and defending the choice with multiple arguments — not just naming the method.

BY THE END OF THIS SHEET, YOU CAN

- ☒ Explain why overfitting produces abnormally large coefficients and why regularization is needed.
- ☒ Write the Ridge loss function ($L_{\text{OLS}} + \lambda m^2$), explain the bias–variance compromise, and derive the 2D closed-form solution.
- ☒ Describe six key properties of Ridge: never-zero coefficients, asymmetric shrinkage, bias–variance impact, geometric shifting, circle constraint ("Ridge" name), and practical heuristic.
- ☒ Write the Lasso loss function and explain the key differences from Ridge: feature selection, sparsity, and when to use each.
- ☒ Explain the geometric intuition: "V"-shaped loss, diamond constraint with sharp corners vs. Ridge's smooth parabola and circular constraint.
- ☒ Prove mathematically why L1 creates sparsity (numerator) while L2 does not (denominator).
- ☒ Select λ via cross-validation, explain standardization, and decide which method suits a given problem.

Worksheet 12 · AML · Lecture 12: Regularization (L1 and L2).